InProject

PORTFOLIO CASE STUDY

Predictive Construction Project Overrun Model

Early-warning classification and regression for cost overruns and schedule delays

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Analytics lifecycle: Ask · Prepare · Process · Analyze · Share · Act

August 2026 · 2,362 projects · 40 predictors · 2019–2025

Synthetic-data disclosure All project records and model outcomes are fictional. Model performance is not an industry benchmark, and production use on real construction projects is not authorized by this portfolio project.

Contents

1. Executive Summary

This case study is the third project in a construction analytics portfolio. It combines the cost and schedule controls developed in Project 1 with the RFI and change-workflow indicators developed in Project 2 to predict future cost overruns and schedule delays from early and mid-project snapshots.

Dataset	Cost model	Schedule model	Test risk bands
2,362 clean projects 1,577 train 349 validation 436 test	Random Forest ROC-AUC 0.899 PR-AUC 0.774	Logistic Regression ROC-AUC 0.756 PR-AUC 0.524	188 Red 138 Yellow 110 Green

Principal conclusions

- Cost-overrun classification generalized strongly to the future synthetic test period.
- Schedule-delay performance decreased from validation to test, creating a real model-monitoring concern rather than a result to hide.
- CPI and current cost variance were the strongest cost-overrun predictors.
- SPI, planned completion, procurement delay, and overdue workflow indicators were important schedule predictors.
- Predictions were converted into Red, Yellow, and Green review priorities, but every score requires human review.

- The Act phase defines a controlled-pilot path; it does not authorize production use.

2. Business Problem and Portfolio Progression

Primary business question

Which early and mid-project indicators best predict material construction cost overruns and schedule delays, and how can management prioritize intervention before the outcomes become unavoidable?

Portfolio progression

Project	Title	Contribution
Project 1	Construction Project Controls Analytics	CPI, SPI, cost, schedule, contingency, EAC, and early-warning controls
Project 2	Construction Change Order and RFI Analytics	RFI response, change approval, pending exposure, revision loops, and forecast lag
Project 3	Predictive Construction Project Overrun Model	Time-validated probability and magnitude prediction with governance

Project 3 uses a new independent synthetic dataset. It references feature concepts from Projects 1 and 2 without mixing their folders, repositories, or records.

3. Analytical Lifecycle and Governance

Phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Ask	Business Understanding	Define	Business Understanding	Initiating / Planning
Prepare	Data Understanding	Measure	Data Acquisition and Understanding	Planning
Process	Data Preparation	Measure / early Analyze	Data Acquisition and Understanding	Executing / Monitoring
Analyze	Modeling and Evaluation	Analyze	Modeling	Executing / Monitoring
Share	Evaluation	Analyze / improvement design	Customer Acceptance	Stakeholder engagement
Act	Deployment	Improve and Control	Deployment / Acceptance	Executing / Monitoring / Closing

The frameworks are conceptually aligned but are not treated as identical. The workflow is iterative, and quality or governance findings may require returning to earlier phases.

4. Dataset, Feature Lineage, and Quality

The deterministic generator created 2,400 intended projects with snapshots between approximately 35% and 60% planned completion. Raw tables intentionally contained duplicates, category variants, missing critical values, invalid numeric values, invalid dates, target inconsistencies, and orphan relationships.

Control	Result
Raw project rows	2,412
Raw workflow rows	2,417
Raw outcome rows	2,416
Duplicate removals	36
Quarantined records	123
Clean model rows	2,362
Cleaning actions	199
SQLite integrity	ok
Foreign-key violations	0

Feature families

- Project and contract attributes
- CPI, SPI, cost variance, schedule variance, contingency, and productivity
- Procurement, weather, and safety indicators
- RFI response, overdue rates, revisions, reopening, and RFI-to-change conversion
- Approved and pending change exposure, approval cycle, old pending rate, and forecast incorporation lag
- Workflow handoffs, revision loops, and snapshot workflow-risk score

Final outcomes, final CPI/SPI, actual end date, final EAC, and final status were excluded from predictor matrices.

5. Modeling Methodology

Element	Method
Time split	Train 2019–2023; Validation 2024; Test 2025
Cost classification	Final cost overrun ≥10%

Element	Method
Schedule classification	Final schedule delay ≥30 days
Cost regression	Final cost-overrun percentage
Schedule regression	Final schedule-delay days
Classification candidates	Logistic Regression and Random Forest
Regression candidates	Ridge Regression and Random Forest Regressor
Champion selection	Validation PR-AUC for classification; validation MAE for regression
Threshold selection	Maximum validation F1; test untouched
Explainability	Permutation importance on held-out test data

The model uses 40 predictors: 31 numeric and 9 categorical. Numeric fields are standardized for linear models, and categorical values are one-hot encoded with unknown-category handling.

6. Classification Performance

Target	Champion	ROC-AUC	PR-AUC	Precision	Recall	F1	Brier
Cost overrun	Random Forest	0.899	0.774	0.625	0.811	0.706	0.123
Schedule delay	Logistic Regression	0.756	0.524	0.520	0.453	0.484	0.183

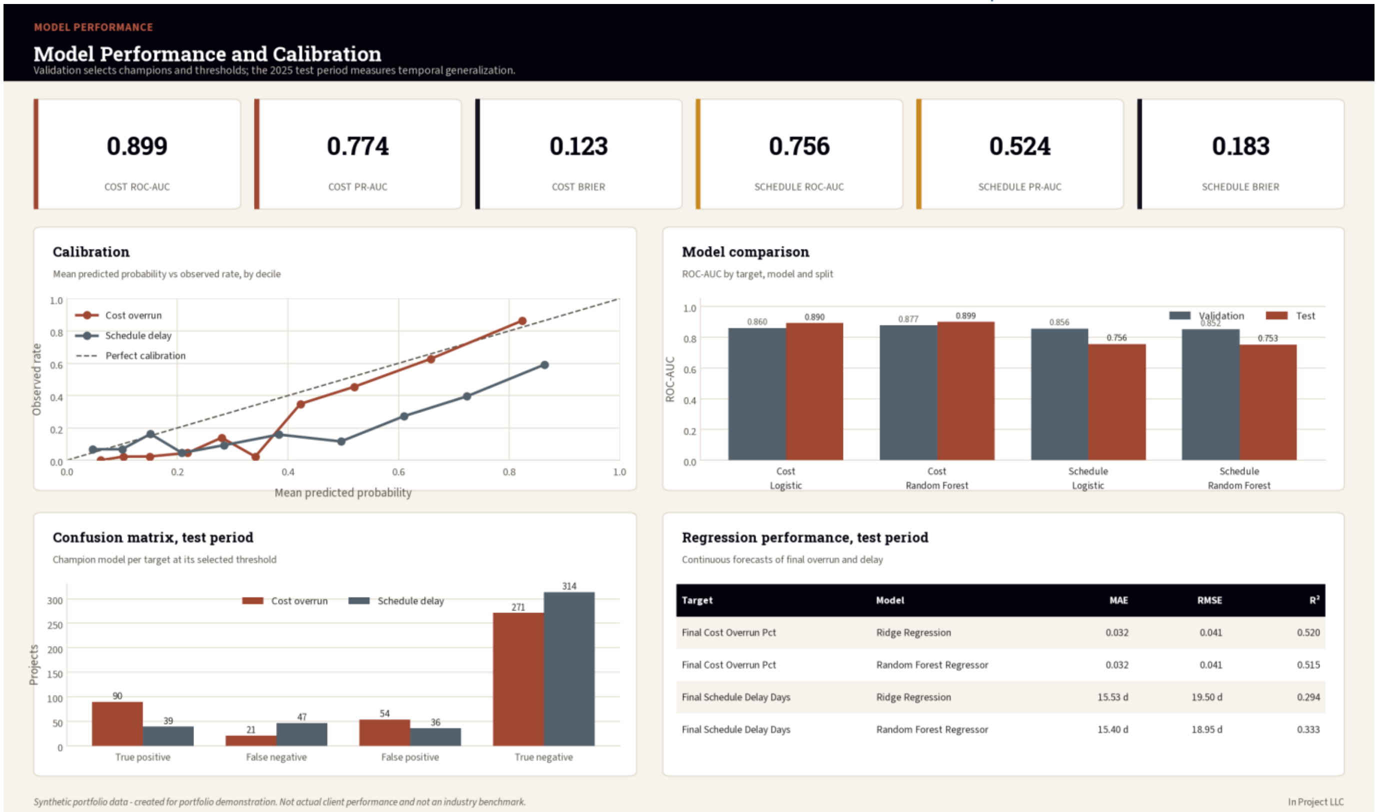


Figure 1. *Model performance and calibration: champion selection on validation, temporal generalization measured on the 2025 test period.*

7. Regression, Calibration, and Temporal Generalization

Target	Model	MAE	RMSE	R^2	Interpretation
Final_Cost_Overrun_Pct	Ridge Regression	0.032	0.041	0.520	Cost-overrun percentage points
Final_Cost_Overrun_Pct	Random Forest Regressor	0.032	0.041	0.515	Cost-overrun percentage points
Final_Schedule_Delay_Days	Ridge Regression	15.528	19.499	0.294	Schedule-delay days
Final_Schedule_Delay_Days	Random Forest Regressor	15.403	18.953	0.333	Schedule-delay days

Temporal degradation

The schedule classifier achieved validation ROC-AUC 0.856 but test ROC-AUC 0.756. The reduction suggests temporal variation in the synthetic portfolio. The result is retained as an example of why model performance must be monitored after deployment.

The monitoring demonstration produced a schedule probability PSI of 0.105, above the proposed Yellow threshold of 0.10.

8. Feature Drivers and Project Risk Scoring

Rank	Cost feature	Importance	Schedule feature
1	CPI_Snapshot	0.0502	SPI_Snapshot
2	Cost_Variance_Pct_Snapshot	0.0289	Planned_Percent_Complete
3	Approved_Change_Value_Pct_Budget	0.0080	Procurement_Delay_Days_to_Date
4	Pending_Change_Exposure_Pct_Budget	0.0030	Overdue_RFI_Rate
5	Labor_Productivity_Index	0.0026	Old_Pending_Change_Rate
6	Contingency_Utilization_Pct	0.0022	Actual_Percent_Complete
7	Procurement_Delay_Days_to_Date	0.0020	RFI_On_Time_Rate
8	SPI_Snapshot	0.0013	Revision_Loop_Rate
9	Snapshot_Data_Completeness_Pct	0.0012	Complexity_Level
10	Complexity_Level	0.0010	Project_Type

Feature importance measures predictive contribution under permutation. It does not establish that changing a feature will cause a better outcome.

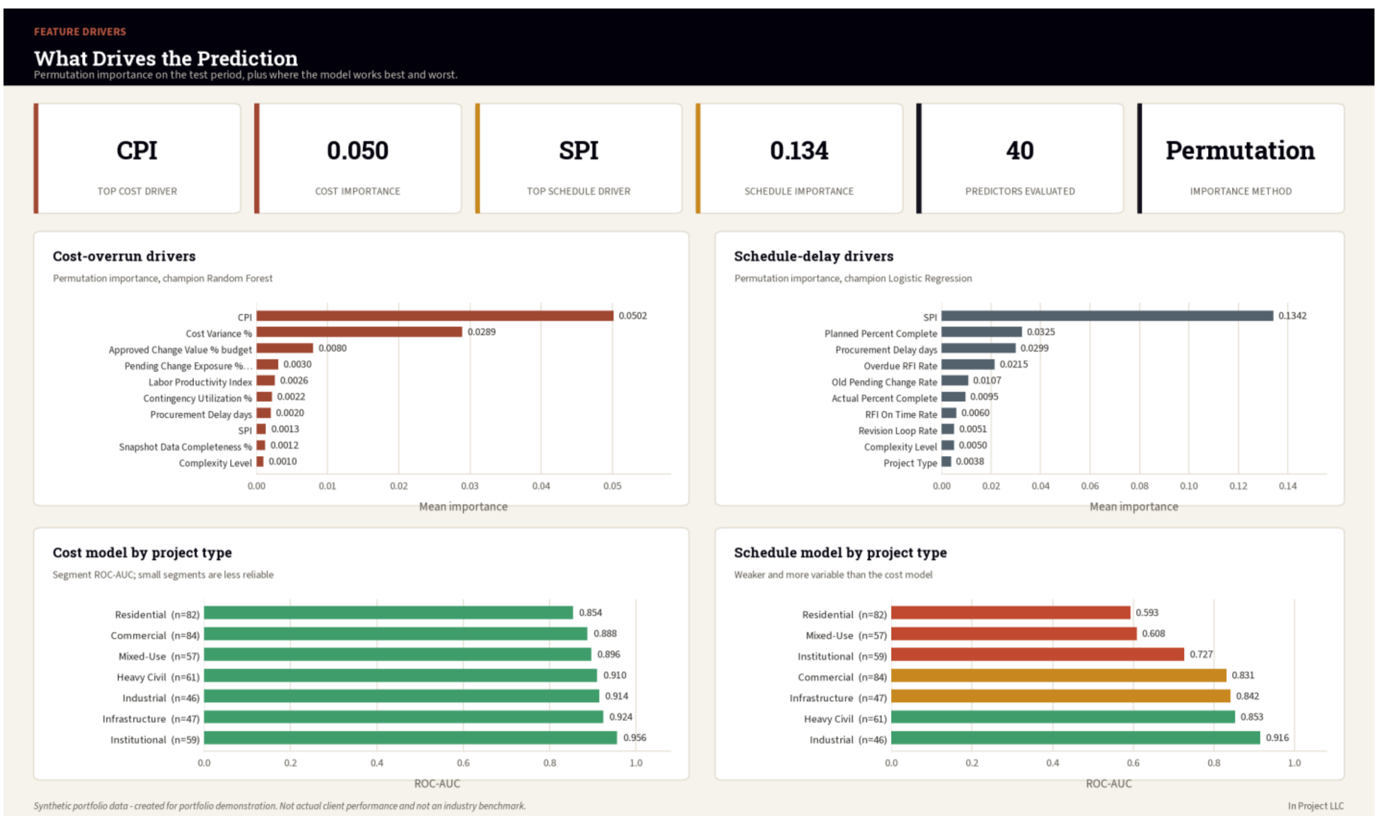


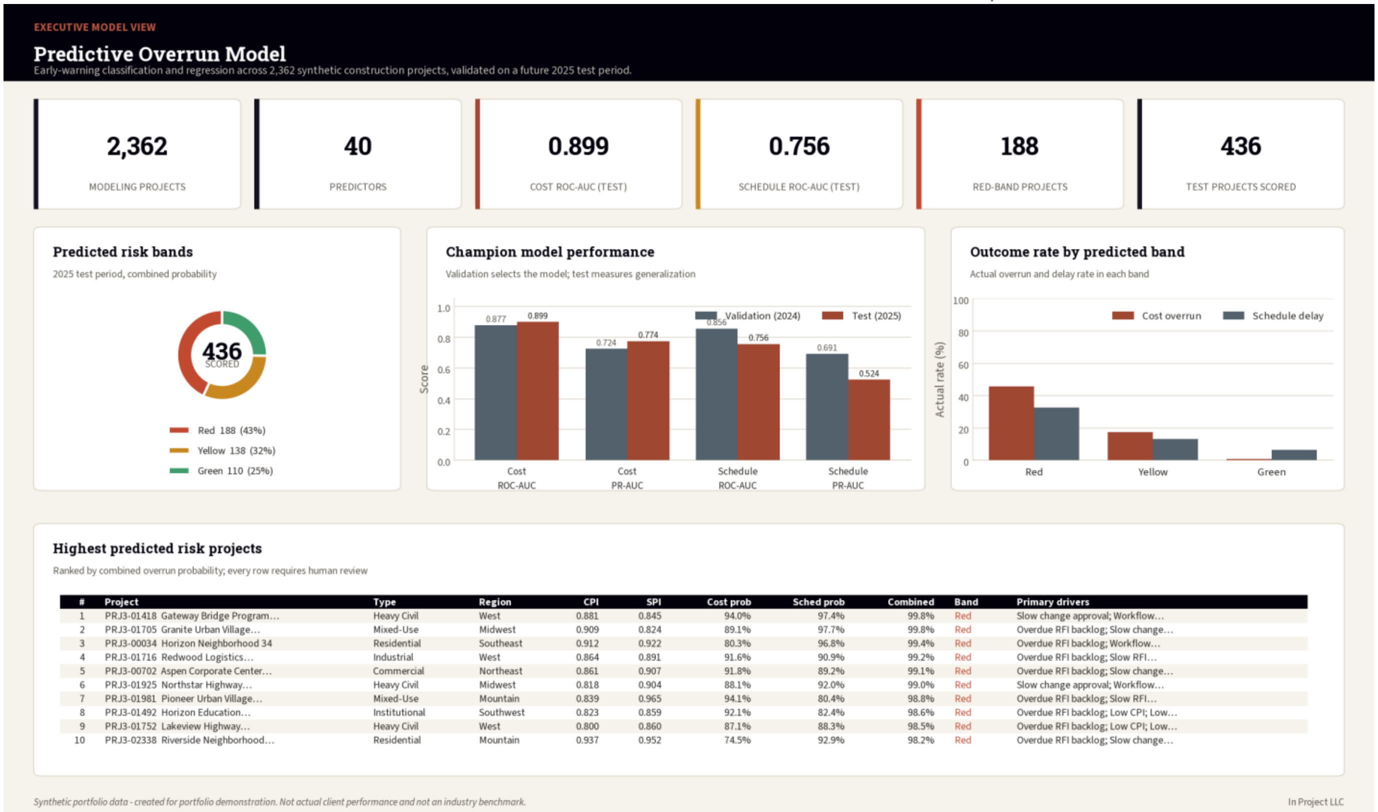
Figure 2. *Permutation importance for both targets, with segment performance showing where each model is weakest.*

Top predicted-risk projects

Project	Band	Overall	Cost	Schedule	Primary drivers
Gateway Bridge Program 1418	Red	99.8%	94.0%	97.4%	Slow change approval; Workflow risk score; Overdue RFI backlog
Granite Urban Village 1705	Red	99.8%	89.1%	97.7%	Overdue RFI backlog; Slow change approval; Workflow risk score
Horizon Neighborhood 34	Red	99.4%	80.3%	96.8%	Overdue RFI backlog; Workflow risk score; Slow RFI response
Redwood Logistics Center 1716	Red	99.2%	91.6%	90.9%	Overdue RFI backlog; Slow RFI response; Workflow risk score
Aspen Corporate Center 702	Red	99.1%	91.8%	89.2%	Overdue RFI backlog; Slow change approval; Slow RFI response
Northstar Highway Improvements 1925	Red	99.0%	88.1%	92.0%	Slow change approval; Workflow risk score; Slow RFI response
Pioneer Urban Village 1981	Red	98.8%	94.1%	80.4%	Overdue RFI backlog; Slow RFI response; Workflow risk score

9. Dashboards and Portfolio Communication

The Share phase produced executive, model-performance, project-risk, and feature-driver dashboards plus Power BI and Tableau build specifications.



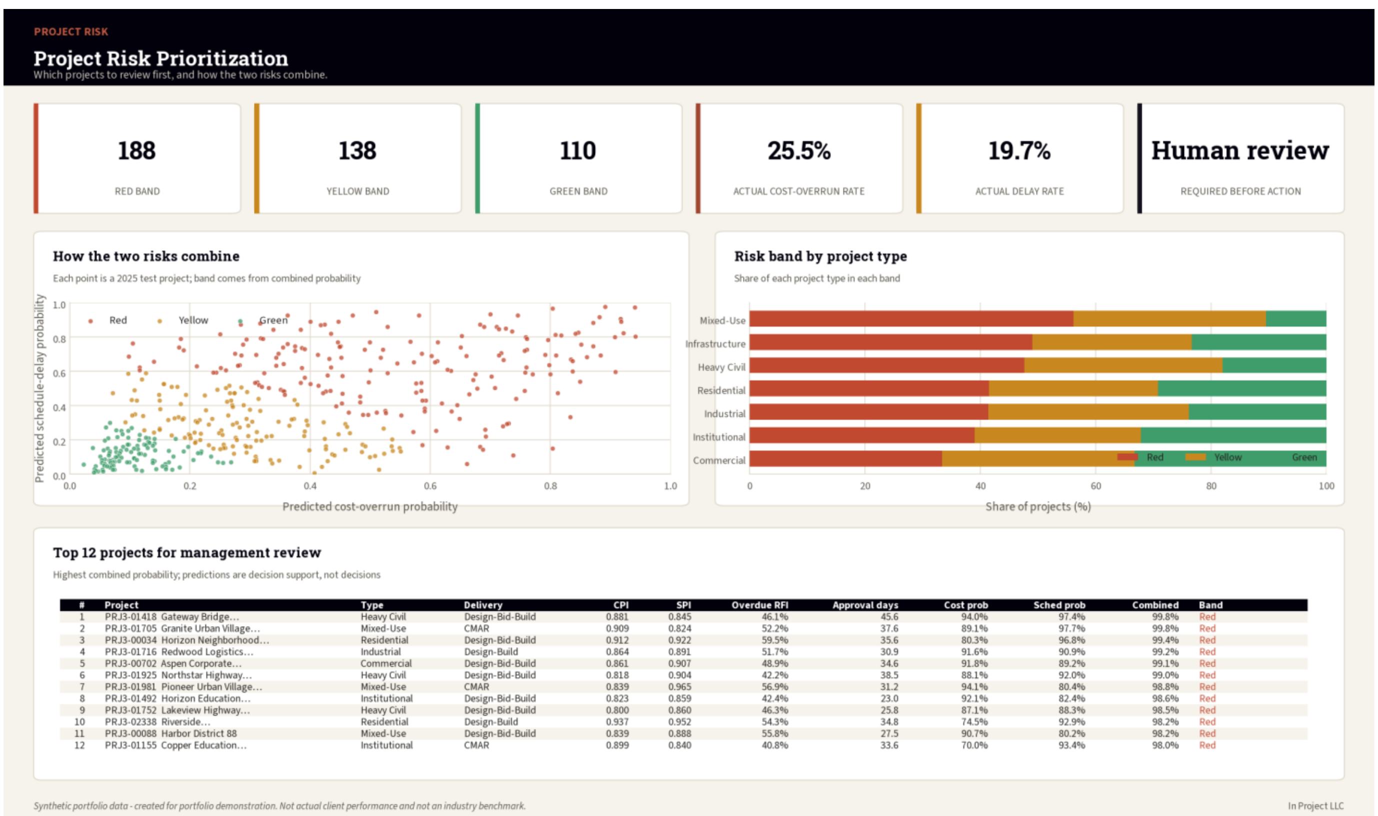
Highest predicted risk projects

Ranked by combined overrun probability; every row requires human review

#	Project	Type	Region	CPI	SPI	Cost prob	Sched prob	Combined	Band	Primary drivers
1	PRJ3-01418 Gateway Bridge Program…	Heavy Civil	West	0.881	0.845	94.0%	97.4%	99.8%	Red	Slow change approval; Workflow…
2	PRJ3-01705 Granite Urban Village…	Mixed-Use	Midwest	0.909	0.824	89.1%	97.7%	99.8%	Red	Overdue RFI backlog; Slow change…
3	PRJ3-00034 Horizon Neighborhood 34	Residential	Southeast	0.912	0.922	80.3%	96.8%	99.4%	Red	Overdue RFI backlog; Workflow…
4	PRJ3-01716 Redwood Logistics…	Industrial	West	0.864	0.891	91.6%	90.9%	99.2%	Red	Overdue RFI backlog; Slow RFI…
5	PRJ3-00702 Aspen Corporate Center…	Commercial	Northeast	0.861	0.907	91.8%	89.2%	99.1%	Red	Overdue RFI backlog; Slow change…
6	PRJ3-01925 Northstar Highway…	Heavy Civil	Midwest	0.818	0.904	88.1%	92.0%	99.0%	Red	Slow change approval; Workflow…
7	PRJ3-01981 Pioneer Urban Village…	Mixed-Use	Mountain	0.839	0.965	94.1%	80.4%	98.8%	Red	Overdue RFI backlog; Slow RFI…
8	PRJ3-01492 Horizon Education…	Institutional	Southwest	0.823	0.859	92.1%	82.4%	98.6%	Red	Overdue RFI backlog; Low CPI; Low…
9	PRJ3-01752 Lakeview Highway…	Heavy Civil	West	0.800	0.860	87.1%	88.3%	98.5%	Red	Overdue RFI backlog; Low CPI; Low…
10	PRJ3-02338 Riverside Neighborhood…	Residential	Mountain	0.937	0.952	74.5%	92.9%	98.2%	Red	Overdue RFI backlog; Slow change…

Synthetic portfolio data - created for portfolio demonstration. Not actual client performance and not an industry benchmark. In Project LLC

Figure 3. *Executive model view: predicted risk bands, champion performance and the highest predicted-risk projects.*



Top 12 projects for management review

Highest combined probability; predictions are decision support, not decisions

#	Project	Type	Delivery	CPI	SPI	Overdue RFI	Approval days	Cost prob	Sched prob	Combined	Band
1	PRJ3-01418 Gateway Bridge…	Heavy Civil	Design-Bid-Build	0.881	0.845	46.1%	45.6	94.0%	97.4%	99.8%	Red
2	PRJ3-01705 Granite Urban Village…	Mixed-Use	CMAR	0.909	0.824	52.2%	37.6	89.1%	97.7%	99.8%	Red
3	PRJ3-00034 Horizon Neighborhood…	Residential	Design-Bid-Build	0.912	0.922	59.5%	35.6	80.3%	96.8%	99.4%	Red
4	PRJ3-01716 Redwood Logistics…	Industrial	Design-Build	0.864	0.891	51.7%	30.9	91.6%	90.9%	99.2%	Red
5	PRJ3-00702 Aspen Corporate…	Commercial	Design-Bid-Build	0.861	0.907	48.9%	34.6	91.8%	89.2%	99.1%	Red
6	PRJ3-01925 Northstar Highway…	Heavy Civil	Design-Bid-Build	0.818	0.904	42.2%	38.5	88.1%	92.0%	99.0%	Red
7	PRJ3-01981 Pioneer Urban Village…	Mixed-Use	CMAR	0.839	0.965	56.9%	31.2	94.1%	80.4%	98.8%	Red
8	PRJ3-01492 Horizon Education…	Institutional	Design-Bid-Build	0.823	0.859	42.4%	23.0	92.1%	82.4%	98.6%	Red
9	PRJ3-01752 Lakeview Highway…	Heavy Civil	Design-Bid-Build	0.800	0.860	46.3%	25.8	87.1%	88.3%	98.5%	Red
10	PRJ3-02338 Riverside…	Residential	Design-Build	0.937	0.952	54.3%	34.8	74.5%	92.9%	98.2%	Red
11	PRJ3-00088 Harbor District 88	Mixed-Use	Design-Bid-Build	0.839	0.888	55.8%	27.5	90.7%	80.2%	98.2%	Red
12	PRJ3-01155 Copper Education…	Institutional	CMAR	0.899	0.840	40.8%	33.6	70.0%	93.4%	98.0%	Red

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Figure 4. *Project risk prioritization: how cost and schedule probability combine into a single review queue.*

The communication design shows probability and risk band together, labels validation and test periods, and keeps synthetic-data, uncertainty, and human-review warnings visible.

10. Act Plan, Human Review, and Monitoring

Controlled-pilot roadmap

- Days 0–30: approve charter, prohibited uses, independent review, reproducibility, source mapping, and data-quality gates.
- Days 31–60: conduct a silent pilot, train reviewers, and review thresholds and monitoring.
- Days 61–90: conduct a limited visible pilot, record decisions and overrides, and make a formal go/no-go decision.

Human review

1. Verify input data and snapshot context.
2. Review cost, schedule, and overall probabilities.
3. Review model and rule-based drivers.
4. Consider current project context and known events.
5. Accept, Override, Escalate, or Defer.
6. Assign an action, owner, due date, and evidence.
7. Record later outcomes for monitoring.

Proposed monitoring

Metric	Yellow	Red / stop
Cost ROC-AUC	Yellow <0.82	Red <0.78
Schedule ROC-AUC	Yellow <0.75	Red <0.70
Probability PSI	Yellow >0.10	Red >0.20
Critical data quality	Any failure	Stop scoring
Segment AUC gap	Yellow >0.10	Red >0.15
Override rate	Yellow >30%	Red >50%
Model age	Review at 12 months	Mandatory decision at 18 months

11. Limitations, Responsible AI, and Conclusion

- All data is synthetic and may not represent real project distributions.
- Performance is not an industry benchmark.
- Target thresholds are case-study definitions.

- Feature importance is predictive, not causal.
- The schedule model shows temporal degradation.
- Segment results have limited sample sizes.
- Real deployment requires approved data, independent validation, calibration, drift review, human oversight, access controls, and incident response.
- The model may not be used for contractual, legal, safety-critical, or personnel decisions.

Conclusion

Project 3 completes a three-project construction analytics portfolio progression from descriptive project controls, to diagnostic workflow analytics, to predictive early warning. The project demonstrates technical modeling and business governance together, including the discipline to retain unfavorable temporal results and restrict unvalidated production use.

12. Sources and Deliverables

Professional references

IBM CRISP-DM overview: https://www.ibm.com/docs/en/spss-modeler/18.6.0?topic=dm-crisp-help-overview

ASQ DMAIC: https://asq.org/quality-resources/dmaic

Microsoft Data Science Lifecycle: https://learn.microsoft.com/en-us/shows/dev-intro-to-data-science/what-is-the-data-science-lifecycle-2-of-28

PMI Process Groups: https://www.pmi.org/standards/process-groups

Primary project deliverables

- Analysis workbook and model-performance tables
- Saved classification and regression pipelines
- Feature manifest and model card
- Executive and operational dashboards
- Power BI and Tableau specifications
- Batch scoring and model-monitoring scripts
- Controlled-pilot Act plan, risk register, RACI, and override log
- Complete Ask–Prepare–Process–Analyze–Share–Act documentation

End of Report